Module-1

1) Define Data Science and Explain its Life cycle.

Definition: Data science is an interdisciplinary field that combines statistics, mathematics, Programming, machine Learning and domain knowledge to collect, Proces analyze and interpret data to extract useful insights and support decision-making.

Life cycle:

1. Problem Definition
   - Understand the business Problem and objectives.
   - Ex: Predict customer churn.
2. Data collection
   - Gather data from databases, APIs, surveys, sensors, etc.

   Ex; collect customer transaction data.
3. Data Preprocessing
   - clean and prepare the data by handling missing values.

   Ex: Fill mising age values.

4. Exploratory Data Analysis (EDA)

- Analyze data using graphs and statistics to discover Patterns.

Ex : Plot customer Purchase trends.

5. Model Building

- Select and train machine learning models.

Ex: Train a Decision Tree classifier.

6. Model Evaluation

- Evaluate model Performance using metrics like accuracy, Precision & recall.

Ex : Accuracy = 95%.

7. Deployment

- Deploy the model into a real-world application

Ex: Fraud detection system in banking.

8. Monitoring & Maintenance

- Monitor Performance & retrain the model when necessary.

Ex: Update the model with new customer data.

2) Differentiate Descriptive, Diagnostic Predictive and Prescriptive Analytics.

Analytics type	Purpose	Questions Answered	Example
Descriptive Analytics	Summarizes historical data	What happened?	Monthly sales report
Diagnostic Analytics	Finds the reason behind an event	Why did it happen?	Sales dropped due to fewer customers
Predictive Analytics	Predicts future outcomes	What is likely to happen?	Forecast next month's sales
Prescriptive Analytics	Recommends the best action	What should we do?	Suggest marketing strategies to increase sales

Examples:

- Descriptive : Total revenue last year.
- Diagnostic : Revenue decreased because of low demand.
- Predictive : Predict next year's revenue.
- Prescriptive : Recommend discounts to improve revenue.

2) Explain the steps Involved in Data Preprocessing with Example.

Data Preprocessing is the Process of cleaning and transforming raw data into a suitable format for analysis.

Steps:

1. Data cleaning
   - Remove duplicate records.
   - Handle missing values.
   - Correct inconsistent data.

Ex: Age = NULL
Replace with average age = 25

2. Data Integration
   - Combine data from multiple sources.

Ex: Merge customer information from sales and marketing databases.

3. Data Transformation
   - convert data into a suitable format.
   - Normalize or standardize values.

Ex: Convert salary from ₹50,000 to 0.5 using normalization.

4. Data Reduction

- Reduce data size while Preserving useful information.

Methods:

- Feature selection
- Sampling
- Dimensionality Reduction

Ex: Select only Age, Income & Purchase columns from a dataset.

5. Data Discretization

- Convert continous values into categories.

Ex: Age:

0-18 → child

19-35 → Adult

36-60 → Middle Age

60+ → senior citizen.